A Case Study Report on

"COMPARATIVE ANALYSIS OF CLASSIFICATION ALGORITHMS FOR BANK MARKETING PREDICTION"

for the course

Fundamentals Of Data Science (22CSC59)

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1. PROBLEM STATEMENT

A bank conducts telephone marketing campaigns to promote term deposits. The purpose of this case study is to predict whether a customer subscribes to a term deposit and to compare four classification algorithms: Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), and Random Forest. The target column **y** has two classes: **yes** (subscribed) and **no** (did not subscribe). This is a binary classification problem.

1.1 Objectives

- Inspect the dataset structure, variable types, missing values, duplicates, and descriptive statistics.
- Explore target-class balance and relationships between customer characteristics and subscription.
- Prepare numeric and categorical features consistently for modelling.
- Train and evaluate four classifiers using the same stratified train-test split.
- Compare accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, and ROC curves.
- Recommend a model with awareness of campaign costs, class imbalance, and prediction-time limitations

2. METHODOLOGY

The workflow is: dataset loading → data-quality checks → exploratory data analysis → train/test split → preprocessing → model training → evaluation and comparison → recommendations. Pandas is used for data handling, Matplotlib for visualizations, and scikit-learn for preprocessing and classification.

2.1 Dataset Description

The supplied **bank-full.csv** dataset contains **45,211 rows and 17 columns**. It includes age, job, marital status, education, default status, account balance, housing/personal loans, contact channel, contact date, call duration, campaign contacts, previous-campaign information, and the target y. The CSV uses a semicolon delimiter. There are 39,922 "no" records and 5,289 "yes" records.

2.2 Data Quality and Feature Selection

The inspected dataset contains no null cells and no duplicate rows. The target is encoded as 1 for "yes" and 0 for "no". The **duration** column is deliberately excluded from the primary models because call duration is only known after the call has taken place; using it for a pre-contact prediction would leak information that is unavailable when deciding whom to call.

2.3 Preprocessing

A stratified 80:20 split with random_state=42 creates 36,168 training records and 9,043 test records. Numerical variables are median-imputed and standardized. Categorical variables are imputed using the most frequent value and one-hot encoded. Preprocessing is fitted within a Pipeline/ColumnTransformer using training data only, then applied to the test data. The same split and preprocessing design are used across the four classifiers.

3. EXPLORATORY DATA ANALYSIS

Exploratory analysis helps describe the data before training models. Graphs are labeled and interpreted below; the results describe patterns in historical records and do not establish causation.

3.1 Target Distribution

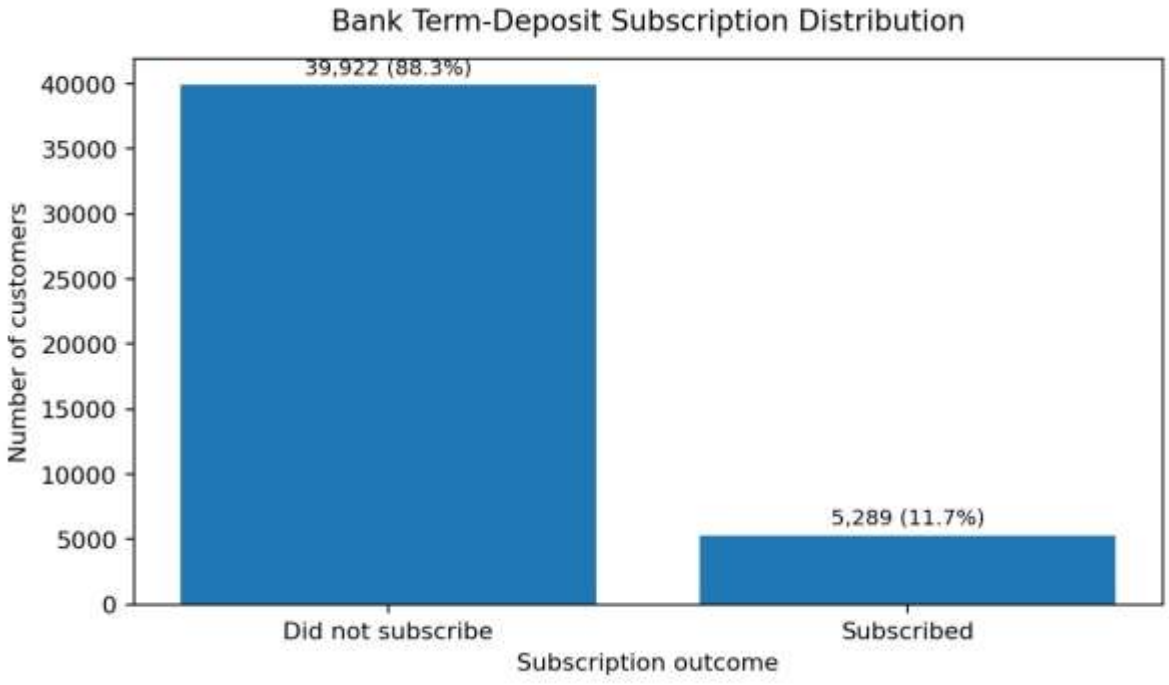


Figure 1. Distribution of term-deposit subscription outcome.

The target is imbalanced: **88.3% (39,922)** of customers did not subscribe, while **11.7% (5,289)** subscribed. A classifier can achieve high accuracy by mostly predicting the majority class while missing potential subscribers. Therefore, accuracy alone is not sufficient for model selection; precision, recall, F1-score, and ROC-AUC must also be considered.

3.2 Customer and Campaign Variables

The dataset contains numeric features such as age, balance, day, campaign, pdays, and previous, along with categorical features such as job, marital status, education, contact, month, and previous campaign outcome. Several categorical fields use "unknown" as a recorded category rather than a null value, so they are retained as categories during one-hot encoding. Balance and campaign variables can be skewed, and extreme values may affect some algorithms.

3.3 Implications for Modelling

Because the positive class is relatively rare, the models are compared on the same held-out test set. Logistic Regression, Decision Tree, and Random Forest use class weighting to address imbalance; KNN is evaluated with distance weighting. These choices do not guarantee that the model will find every subscriber, so recall and precision are examined together.

4. MODEL BUILDING AND EVALUATION

Four classifiers were trained with a shared preprocessing pipeline. The test set contains 9,043 customers, including 1,058 subscribers and 7,985 non-subscribers. Metrics use "yes"/subscription as the positive class.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Random Forest	0.8663	0.4412	0.5359	0.4840	0.8027
Decision Tree	0.8122	0.3319	0.5974	0.4267	0.7764
Logistic Regression	0.7548	0.2662	0.6238	0.3732	0.7722
KNN	0.8926	0.6516	0.1767	0.2781	0.7369

4.1 Interpretation of Metrics

Accuracy measures the share of all predictions that are correct. **Precision** measures how many predicted subscribers actually subscribed. **Recall** measures the share of actual subscribers identified. **F1-score** balances precision and recall. **ROC-AUC** measures how well a model ranks positive cases above negative cases across thresholds. With an imbalanced target, F1-score and recall are particularly informative for campaign targeting.

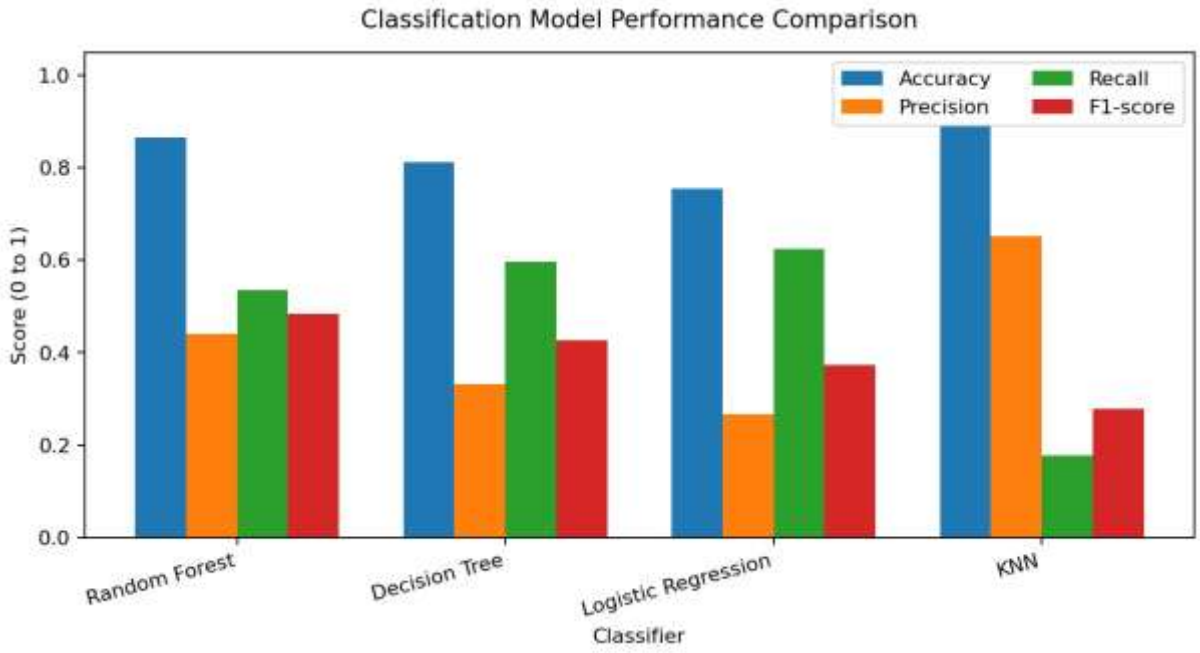


Figure 2. Comparison of classification metrics across the four models.

Random Forest achieved the highest F1-score (0.4840) and ROC-AUC (0.8027), indicating the strongest overall balance among these tested models. KNN had the highest accuracy (0.8926) and precision (0.6516), but recall was only 0.1767:
it identified fewer than one in five actual subscribers. The best model therefore depends on whether the campaign prioritizes finding more subscribers or limiting unnecessary contacts.

4.2 Random Forest Confusion Matrix

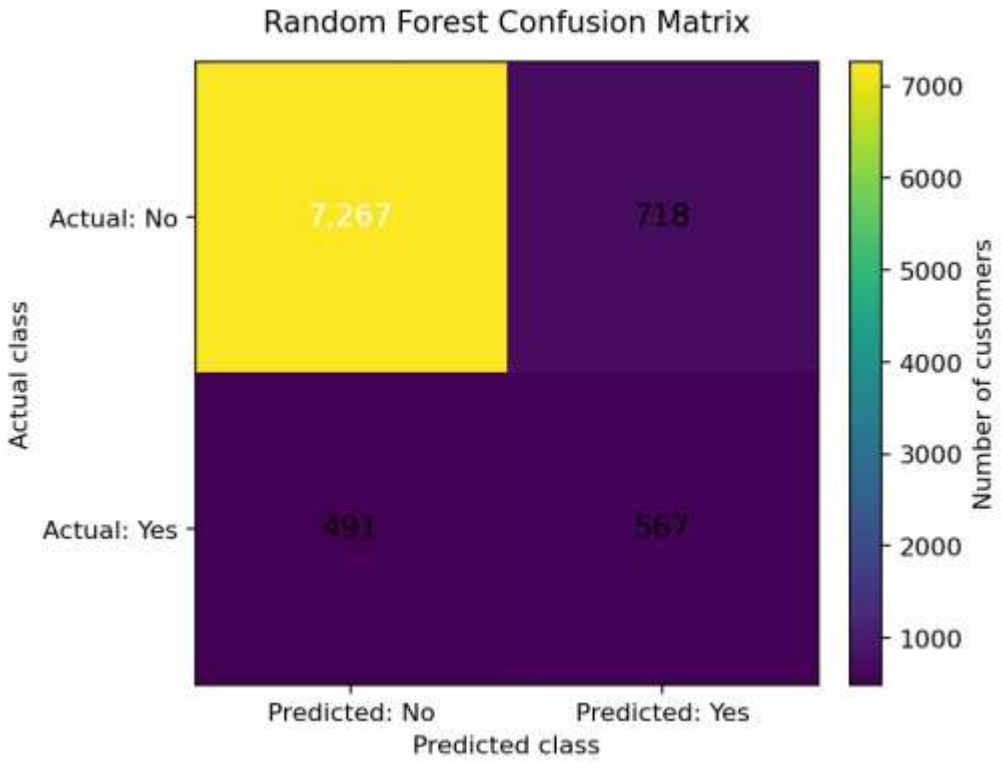


Figure 3. Confusion matrix for the selected Random Forest model.

For the selected Random Forest model, the confusion matrix is **[[7267, 718], [491, 567]]**, where rows are actual classes (no, yes) and columns are predicted classes (no, yes). The model correctly classified 7,267 non-subscribers and 567 subscribers. It incorrectly predicted subscription for 718 customers who did not subscribe (false positives) and missed 491 actual subscribers (false negatives). Thus, it identified 567 of 1,058 actual subscribers, consistent with recall of about 53.6%.

4.3 ROC-AUC Comparison

The Random Forest ROC-AUC is 0.8027, followed by Decision Tree (0.7764), Logistic Regression (0.7722), and KNN (0.7369). ROC-AUC summarizes ranking performance across probability thresholds; higher values indicate stronger separation. The final operating threshold should be selected using campaign capacity and the relative cost of missed subscribers versus contacting non-subscribers.

5. STATISTICAL FINDINGS

The dataset includes 45,211 customer records and 17 columns. The target variable is strongly imbalanced, with 5,289 subscribers (11.7%) and 39,922 non-subscribers (88.3%). The inspected data has no missing cells and no duplicate rows. These findings establish that the main data-preparation concern is not null-value removal, but handling mixed feature types and a minority positive class.

The four models show different trade-offs. KNN provides the highest accuracy and precision but has the lowest recall, meaning it misses many actual subscribers. Logistic Regression has the highest recall among the tested models (0.6238), but its precision is low (0.2662). Decision Tree provides intermediate results. Random Forest has the highest F1-score and ROC-AUC, giving the most balanced result by the selected criteria.

6. RECOMMENDATIONS

- Use Random Forest as the initial candidate when balancing precision and recall with F1-score and ROC-AUC, butvalidate it on newer campaign data before operational use.
- Rank customers by predicted subscription probability and choose a contact threshold that fits campaign capacity andbusiness costs.
- If the priority is to identify as many potential subscribers as possible, compare the higher-recall Logistic Regressionmodel and assess the extra contacts it generates.
- Monitor precision, recall, F1-score, and ROC-AUC over time; accuracy alone can hide poor performance on theminority class.
- Do not use duration for pre-contact predictions. If a separate post-call analysis is performed, clearly label it as adifferent prediction setting.

6.1 Limitations

The results are based on one stratified random train-test split and historical data, so performance may change across time, regions, and campaign strategies. “Unknown” values may contain useful information but may also reflect data-collection practices. The analysis does not establish causal effects. The business costs of false positives and false negatives were not supplied, so no optimal decision threshold can be claimed.

7. CONCLUSION

This case study compared Logistic Regression, Decision Tree, KNN, and Random Forest for predicting bank term-deposit subscription. The Bank Marketing dataset contains 45,211 records, and the target is imbalanced, with only 11.7% positive cases. A shared preprocessing pipeline and stratified 80:20 split were used to make the comparison consistent, while the duration field was excluded to support realistic pre-contact prediction.

Random Forest achieved the highest F1-score (0.4840) and ROC-AUC (0.8027). KNN had the highest accuracy (0.8926) but a low recall (0.1767), demonstrating why accuracy alone is not enough. Random Forest is the strongest overall candidate under the selected F1-score criterion, while Logistic Regression may be worth considering when identifying more subscribers is the priority. Final deployment decisions should consider campaign capacity, error costs, and validation on recent data.

REFERENCES

- Bank Marketing dataset (bank-full.csv) supplied for the case study; original dataset from the UCI Machine LearningRepository.
- Scikit-learn documentation for Logistic Regression, DecisionTreeClassifier, KNeighborsClassifier,RandomForestClassifier, Pipeline, ColumnTransformer, preprocessing, and classification metrics.
- Pandas, NumPy, and Matplotlib documentation for data handling and visualization.